

Privacy-Aware Smart Home Monitoring for Daily Routine Anomaly Detection

Project 35 - Ambient Assisted Living and Routine Monitoring - Tier A - Advanced - Max 100p

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Artifact type: course prototype, dataset protocol, dashboard, evaluation report and IEEE-style documentation support.

Implementation stack: Python, pandas, scikit-learn, Streamlit, public CASAS Aruba smart-home event data.

Abstract

This project implements a privacy-aware smart-home monitoring prototype for detecting daily routine anomalies from ambient sensor data. The system avoids intrusive sensing such as cameras and microphones and uses minimized sensor-derived features instead. The prototype models daily routines, detects deviations such as inactivity, unusual nighttime movement, missed morning routines and missing-data cases, and displays results through a Streamlit dashboard. The evaluation uses the public CASAS Aruba dataset and controlled synthetic anomaly injection because explicit anomaly ground truth is not available in the raw dataset.

1. Assignment Mapping

The following table maps the initial assignment requirements to the implemented project. The network-anonymity baseline paragraph mentioning Tor/VPN appears unrelated to the smart-home theme; in this project it is handled as a privacy baseline comparison between raw-derived, pseudonymized, aggregated and minimized smart-home data.

Initial requirement	How it is covered in the project
Theme: Ambient Assisted Living and Routine Monitoring	Uses smart-home ambient sensor data to model daily routines for one resident.
Research problem	Detect routine anomalies while minimizing sensitive data collection.
Objective	Build an ambient sensing prototype for routine modeling and anomaly detection.
Task description	Detects inactivity, unusual nighttime movement, missed morning routine and missing-data/sensor-failure cases.
Expected technical components	Ambient sensor data, time-series features, rule-based model, Isolation Forest, privacy layer and Streamlit dashboard.
Core requirements	Data minimization, anomaly model, false-positive analysis, caregiver notification policy and ethical review are included.
Deliverables	Prototype code, dataset protocol, dashboard, evaluation report support, README and this IEEE-style documentation.
Assessment focus	Anomaly quality, privacy design, usability and safety escalation logic are explicitly evaluated.

2. Problem Definition

Smart homes can support independent living, but continuous monitoring can become intrusive if raw sensor streams, exact locations, cameras or microphones are collected without limitation. The system therefore detects anomalies from aggregated and minimized routine features rather than from invasive data.

Detected anomaly categories: severe inactivity, unusual nighttime movement, missed morning routine and possible sensor failure or missing data.

3. Three-Week Scope and Work Plan

The project scope is intentionally narrow and testable. It uses a public dataset, reproducible scripts and controlled synthetic anomalies instead of real surveillance or clinical deployment.

Week	Implementation tasks	Expected output
Week 1	Select CASAS Aruba, define input/output, parse CSV/TXT events, normalize timestamps, create daily routine features and document dataset protocol.	Clean event pipeline, features_daily.csv, dataset protocol.
Week 2	Implement rule-based anomaly detector, Isolation Forest, privacy levels P0-P3, synthetic anomaly injection and evaluation metrics.	Predictions, metrics.json, false-positive/false-negative analysis.
Week 3	Build Streamlit dashboard, finalize caregiver notification policy, write ethical review, README and IEEE-style documentation.	Dashboard, final report, reproducible commands and screenshots.

4. Dataset Protocol

The recommended dataset is CASAS Dataset 11 - Aruba. It contains daily-life smart-home events for one resident from 2010-2011 and is annotated. The project also supports similar CSV/TXT CASAS-style event files.

Expected event format

```
2010-11-04,00:03:50.209589,Bedroom,ON
2010-11-04,00:03:57.399391,Bedroom,OFF
2010-11-04,02:32:33.351906,Bedroom,ON
```

The parser supports both headerless CSV files and files with explicit columns such as date, time, sensor_id and message.

5. System Architecture

Stage	Description	Output
Data ingestion	Read CASAS CSV/TXT event stream	Normalized events
Feature extraction	Daily and hourly routine features	features_daily.csv
Privacy layer	Pseudonymization, aggregation, minimization	P0-P3 feature sets
Anomaly detection	Rule-based baseline + Isolation Forest	Anomaly score and label
Notification policy	Map anomalies to escalation levels	Notification level 0-3
Dashboard	Streamlit visualization	Interactive prototype

6. Extracted Features

The model does not need raw event logs during prediction reporting. Instead, each day is summarized through privacy-aware features.

Feature	Meaning
event_count	Total number of smart-home events in one day
motion_count_total	Number of active motion/activity observations
night_motion_count	Activity count between 00:00 and 05:00
morning_motion_count	Activity count between 06:00 and 11:00
longest_inactivity_gap_minutes	Longest gap without detected activity
active_sensors_count	Number of unique active sensors during the day
active_sensor_types_count	Number of broad sensor categories active in the day

7. Privacy-by-Design

The project compares smart-home privacy levels rather than unrelated network tools. The goal is to keep enough information for anomaly detection while avoiding permanent storage of raw, fine-grained behavior traces.

Level	Data representation	Privacy level	Detection utility
P0	Raw-derived daily features	Low	High
P1	Pseudonymized sensor identifiers	Medium	High
P2	Hourly aggregated counts	Good	Medium-High
P3	Daily minimized routine features	Very good	Medium

Data minimization: the dashboard and prediction output use feature summaries rather than raw sensor event streams. **Data leakage control:** y_true and output columns are removed before ML training and prediction.

8. Anomaly Detection Methods

8.1 Rule-based baseline

The rule-based detector computes thresholds from the training period. It flags inactivity, severe inactivity, unusual night movement, missed morning routine and possible sensor failure/missing-data patterns. It is interpretable and suitable as a conservative safety baseline.

8.2 Isolation Forest

The Isolation Forest model is an unsupervised anomaly detector trained on normal daily feature vectors. It provides an additional anomaly score for decision support. It is not used alone for critical escalation because it may miss some

safety-relevant anomalies.

9. Evaluation Design and Success Criteria

The Aruba dataset provides routine activity events, but not direct binary anomaly labels. Therefore, the evaluation injects controlled synthetic anomalies only into the test portion of the dataset. This keeps the experiment reproducible and ethically safe.

Scenario	Injected behavior	Expected notification
Severe inactivity	Very low activity and long inactivity gap	Level 3
Unusual nighttime movement	High night_motion_count	Level 1
Missed morning routine	morning_motion_count set to zero	Level 1-2
Sensor failure / failure case	Zero events and zero active sensors	Level 3

Success criteria: the prototype must run reproducibly, show routine features in the dashboard, detect injected anomalies, report false positives/false negatives and avoid medical or production-ready claims.

10. Results

The following results were obtained after removing `y_true` from the ML feature set. Therefore, the Isolation Forest model no longer sees the correct answer during training or prediction.

Model	Accuracy	Precision	Recall	F1	TN	FP	FN	TP
Rule-based	0.818	0.231	1.000	0.375	168	40	0	12
Isolation Forest	0.877	0.222	0.500	0.308	187	21	6	6

The rule-based detector achieved perfect recall on the synthetic anomaly scenarios, detecting all injected anomalies, but produced more false positives. The Isolation Forest model achieved higher overall accuracy and fewer false positives, but missed several injected anomalies. Therefore, rule-based detection is used for safety-critical escalation and the ML anomaly score is treated as decision support.

11. Caregiver Notification Policy

Level	Meaning	Action
0	Normal	No notification
1	Minor anomaly	Soft alert in dashboard
2	Repeated/moderate anomaly	Optional caregiver notification
3	Severe inactivity or missing-data pattern	Urgent caregiver notification

The system reports routine deviations, not diagnoses. A correct message is: "Unusual inactivity detected compared to the resident normal routine." The system must not claim: "The resident has fallen."

12. Safety, Privacy and Ethical Review

The project uses public smart-home data and controlled synthetic anomalies. It does not monitor third parties, does not bypass institutional policies, does not use cameras or microphones and does not attempt to infer sensitive personal attributes. It is a course prototype, not a production-ready or clinically validated monitoring system.

13. Deliverables

Deliverable	Included artifact
Prototype	Python scripts in <code>src/</code> and <code>scripts/</code>
Dataset protocol	<code>reports/dataset_protocol.md</code> and Section 4 of this PDF
Dashboard	<code>dashboard/app.py</code> , launched with <code>streamlit</code>
Evaluation report	<code>metrics.json</code> , <code>predictions_combined.csv</code> and Sections 9-10
IEEE-style paper support	<code>reports/ieee_paper_template.md</code> and this structured PDF

14. Reproducible Commands

```
source .venv/bin/activate.fish
rm -rf outputs/* data/processed/*
python scripts/01_prepare_dataset.py --input data/raw/aruba.csv
```

```
python scripts/04_make_synthetic_labels.py  
python scripts/02_train_evaluate.py  
python -m streamlit run dashboard/app.py
```